Searching Tables.

Creating Pre-filled Tables

A B C D E F G H I J K L

```
01 LetterTable.
   02 TableValues.
      03 FILLER PIC X(13)
         VALUE "ABCDEFGHIJKLM".
      03 FILLER PIC X(13)
         VALUE "NOPQRSTUVWXYZ".
```

Creating Pre-filled Tables

A	B	C	D	E	F	G	H	I	J	K	L

```
01 LetterTable.
   02 TableValues.
      03 FILLER PIC X(13)
         VALUE "ABCDEFGHIJKLM".
      03 FILLER PIC X(13)
         VALUE "NOPQRSTUVWXYZ".

   02 FILLER REDEFINES TableValues.
      03 Letter PIC X OCCURS 26 TIMES.
```

Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L
1	2	3	4	5	6	7	8	9	10	11	12

```
01 LetterTable.
   02 TableValues.
      03 FILLER PIC X(13)
         VALUE "ABCDEFGHIJKLM".
      03 FILLER PIC X(13)
         VALUE "NOPQRSTUVWXYZ".
   02 FILLER REDEFINES TableValues.
      03 Letter PIC X OCCURS 26 TIMES.
```

```
PERFORM VARYING Idx FROM 1 BY 1 UNTIL
   LetterIn EQUAL TO Letter(Idx)
END-PERFORM.
DISPLAY LetterIn, "is in position ", Idx.
```

Search Syntax

$$\underline{\text{SEARCH}}\quad \text{TableName}\left[\underline{\text{VARYING}}\begin{Bmatrix}\text{Indentifier}\\ \text{IndexName}\end{Bmatrix}\right]$$

$$\left[\text{AT } \underline{\text{END}} \text{ StatementBlock}\right]$$

$$\left\{\underline{\text{WHEN}}\ \text{Condition}\begin{Bmatrix}\text{Statementblock}\\ \underline{\text{NEXT SENTENCE}}\end{Bmatrix}\right\}\ldots$$

$$\underline{\text{END – SEARCH}}$$

$$\underline{\text{OCCUR}}\text{S}\quad \text{TableSize}\quad \text{TIMES}$$

$$\left[\underline{\text{INDEXED}}\ \text{BY}\ \{\text{IndexName}\}\ldots\right]$$

SET Syntax

$$\text{SET} \begin{Bmatrix} \text{IndexName} \\ \text{Identifier} \end{Bmatrix} \ldots \text{TO} \begin{Bmatrix} \text{IndexName} \\ \text{Identifier} \\ \text{Integer} \end{Bmatrix}$$

$$\text{SET} \; \{\text{IndexName}\} \ldots \begin{Bmatrix} \text{UP} \\ \text{DOWN} \end{Bmatrix} \text{BY} \begin{Bmatrix} \text{Identifier} \\ \text{Integer} \end{Bmatrix}$$

$$\text{SET} \; \{\text{ConditionName}\} \ldots \text{TO} \;\; \text{TRUE}$$

Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L
1	2	3	4	5	6	7	8	9	10	11	12

```
01 LetterTable.
   02 TableValues.
      03 FILLER PIC X(13)
         VALUE "ABCDEFGHIJKLM".
      03 FILLER PIC X(13)
         VALUE "NOPQRSTUVWXYZ".
   02 FILLER REDEFINES TableValues.
      03 Letter PIC X OCCURS 26 TIMES
                      INDEXED BY LetterIdx.

SET LetterIdx TO 1.
SEARCH Letter
   AT END DISPLAY "Letter not found!"
   WHEN Letter(LetterIdx) = LetterIn
        DISPLAY LetterIn, "is in position ", Idx
END-SEARCH.
```

Searching a Two Dimension Table.


```
01 TimeTable.
   02 Day OCCURS 5 TIMES INDEXED BY DayIdx.
      03 Hours OCCURS 8 TIMES INDEXED BY HourIdx.
         04 Item        PIC X(10).
         04 Location    PIC X(10).

SET DayIdx TO 0.
PERFORM UNTIL MeetingFound OR DayIdx > 5
   SET DayIdx UP BY 1
   SET HourIdx TO 1
   SEARCH Hours WHEN MeetingType = Item(DayIdx, HourIdx)
      SET MeetingFound TO TRUE
      DISPLAY MeetingType " on " DayIdx " at " HourIdx
   END-SEARCH
END-PERFORM.
```

Search All Syntax.

SEARCH ALL TableName [AT END StatementBlock]

WHEN { ElementIdentifier { IS EQUAL TO / IS = } { Identifier / Literal / ArithExpression } / ConditionName }

[AND { ElementIdentifier { IS EQUAL TO / IS = } { Identifier / Literal / ArithExpression } / ConditionName }]

{ StatementBlock / NEXT SENTENCE }

END – SEARCH

OCCURS TableSize TIMES

[{ ASCENDING / DESCENDING } KEY IS {ElementIdentifier} …] …

[INDEXED BY {IndexName}…]

Using the Search All


```
01 LetterTable.
   02 TableValues.
      03 FILLER PIC X(13)
         VALUE "ABCDEFGHIJKLM".
      03 FILLER PIC X(13)
         VALUE "NOPQRSTUVWXYZ".
   02 FILLER REDEFINES TableValues.
      03 Letter PIC X OCCURS 26 TIMES
                      ASCENDING KEY IS Letter
                      INDEXED BY LetterIdx.

SEARCH ALL Letter
    WHEN Letter(LetterIdx) = LetterIn
        DISPLAY LetterIn, "is in position ", Idx
END-SEARCH.
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
1	26	13	=	M

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
14	26	13	=	M

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
14	26	20	=	T

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
14	19	20	=	T

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
14	19	16	=	P

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
17	19	16	=	P

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
17	19	18	=	R

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
17	17	18	=	R

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
17	17	18	=	Q

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
17	17	18	=	Q

ALGORITHM.

```
Middle = (Lower + Upper) / 2
CASE TRUE
   WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1
   WHEN Letter(Middle) > "Q" THEN Upper = Middle -1
   WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE
   WHEN Lower > Upper THEN ItemNotInTable TO TRUE
```

Search All Example.

```
01 StateTable.
   02 StateValues.
      03 FILLER PIC X(20) VALUE ??????????????
             Post Codes and Names
      03 FILLER PIC X(20) VALUE ??????????????

   02 FILLER REDEFINES StateValues.
      03 States OCCURS 50 TIMES
            ASCENDING KEY IS StateName
            INDEXED BY StateIdx.
         04 PostCode     PIC X(6).
         04 StateName    PIC X(14).

SEARCH ALL States
    AT END DISPLAY "State not found"
    WHEN StateName(StateIdx) = InputName
        MOVE PostCode(StateIdx) TO PrintPostCode
END-SEARCH.
```